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| EDUCATION | <b>University of Illinois at Urbana-Champaign</b><br><i>Ph.D. in Economics</i><br>Advisor: Prof. Tatyana Deryugina | Champaign, IL<br>May 2026 ( <i>expected</i> ) |
|           | <b>University of Illinois at Urbana-Champaign</b><br><i>M.S. in Policy Economics</i>                               | Champaign, IL<br>Dec 2019                     |
|           | <b>University of International Business and Economics (UIBE)</b><br><i>B.S. in Economics (Minor in French)</i>     | Beijing, China<br>Jun 2018                    |
|           | <b>University of California, Berkeley</b><br><i>Exchange student</i>                                               | Berkeley, CA<br>Jan 2017 - May 2017           |

RESEARCH INTEREST      **Causal Inference, Environmental Economics, Behavioral Economics**

WORKING PAPERS      **Air Pollution and Economic Activity: Evidence from Foot Traffic Patterns in the U.S.** (Job Market Paper)

I investigate how air pollution affects economic activity. Using over 600 million phone-location-based foot traffic data points from SafeGraph, I conduct a large-scale analysis to examine the causal effect of air pollution on activity patterns across the US. Using changes in local wind direction as an instrumental variable (IV) for air pollution, I find that a  $1 \mu\text{g}/\text{m}^3$  increase in PM2.5 concentration leads to a 0.50% decrease in economic activity, resulting in a nationwide reduction of 43 million trips annually. The reductions are widespread across different economic sectors, with recreational activities experiencing the largest decline. The effect is more pronounced in higher-income counties and areas with a larger share of children, suggesting greater awareness among wealthier or more vulnerable populations.

**Go with the Wind: Polluters' Strategic Response to Wind Directions**

I investigate the strategic responses of polluters in the US. Under the Clean Air Act, data from air quality monitors are crucial for determining whether an area meets environmental standards. Since air pollution can be easily carried by wind, these monitors capture more pollution from upwind areas and less from downwind areas. As a result, polluters may be incentivized to emit less on days when they are upwind and more when they are downwind. I identify such strategic behavior among US power plants, finding that a one-standard-deviation increase in favorable wind direction (i.e., when the wind blows pollutants away from monitors) leads to a 0.8% and 0.5% increase in sulfur dioxide ( $\text{SO}_2$ ) and nitrogen dioxide ( $\text{NO}_x$ ) emissions, respectively. At the same time, fuel input remains unchanged, but the emission rate rises, suggesting that power plants temporarily turn off pollution control equipment. Additionally, the increase is more pronounced when power plants and monitors are in the same state, suggesting that local governments may be more lenient when local polluters' emissions do not negatively impact their monitoring outcomes.

## WORK IN PROGRESS

### **Natural Disasters and Occupational Mobility: Evidence from the 1927 Mississippi Flood** (joint with Xiaohan Wang)

We investigate how a major natural disaster affect intergenerational occupational mobility in the agricultural sector. Linking individuals across the 1920, 1930, and 1940 Censuses, we show that children residing in counties affected by the 1927 Mississippi Flood were significantly less likely to become farmers as adults. This effect is mainly driven by children from non-farming families, who were 9% less likely to work in agriculture and 10% more likely to transition into the service sector. In the longer run, these individuals achieved socioeconomic status scores about 3% higher than comparable children in non-flooded counties. Our findings suggest that natural disasters can accelerate structural change by shifting new labor market entrants out of agriculture and into higher-status occupations.

### **Social Networks and Household Protective Investments: Evidence from Wildfire Smoke** (joint with Sanjukta Mitra)

### **Deforestation and Air Pollution in the Indonesian Rainforest** (joint with Francesco Cenerini)

## CONFERENCES AND WORKSHOPS

### **2026**

ASSA Annual Meeting (Philadelphia)

### **2025**

South Economics Association Annual Conference (Tampa), program in Environmental and Resource Economics Seminar (UIUC), Midwest Economics Association Annual Conference (Kansas City), East Economics Association Annual Conference (New York City), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

### **2024**

Occasional Workshop in Environmental and Resource Economics (UCSB), Online Summer Workshop in Environment, Energy, and Transportation (Virtual), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

### **2023**

Price Theory Summer Camp (University of Chicago), Applied Micro Research Lunch (UIUC)

### **2022**

Berkeley/Solan Summer School in Environmental and Energy Economics, Applied Micro Research Lunch (UIUC)

## TEACHING EXPERIENCE

### **ECON 102: Microeconomic Principles**

Fall 2021, Spring 2022

Teaching Assistant

### **The Economics of the Firm (EMBA at the University of Warsaw)**

Fall 2019

Teaching Assistant to Prof. Hadi Esfahani

### **Econ 528: Microeconomics for Business**

Spring 2019, Summer 2019

Course Assistant

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| WORKING<br>EXPERIENCE | <b>Gies College of Business, UIUC</b><br>Research Assistant to Prof. Tatyana Deryugina                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Aug 2022 – present   |
|                       | <b>Department of Economics, UIUC</b><br>Research Assistant to Prof. Eunyi Chung                                                                                                                                                                                                                                                                                                                                                                                                                                                              | June 2021 – Aug 2022 |
|                       | <b>China Institute for Educational Finance Research, Peking University</b><br>Research Associate                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mar 2020 – Aug 2020  |
|                       | <b>Risk Management Department, Bank of Communications</b><br>Data Analyst Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jun 2017 – Aug 2017  |
|                       | <b>School of International Trade and Economics, UIBE</b><br>Research Assistant to Prof. Deyu Yuan                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sep 2016 – Jan 2017  |
| AWARDS                | <ul style="list-style-type: none"> <li>• Robert Willis Harbeson Memorial Dissertation Fellowship, University of Illinois 2025</li> <li>• Hung-Chao and Julia Tai Family Fellowship, University of Illinois 2025</li> <li>• Cleo Fitzsimmons Award, University of Illinois 2022</li> <li>• Graduate Fellowship, University of Illinois 2020</li> </ul>                                                                                                                                                                                        |                      |
| SKILLS                | <b>Languages:</b> Chinese (Native), English (Proficient), French (Basic).<br><b>Programming:</b> R, Python, Stata, Git, Shell.                                                                                                                                                                                                                                                                                                                                                                                                               |                      |
| REFERENCES            | <b>Tatyana Deryugina (Chair)</b><br>Associate Professor of Finance<br>Gies College of Business<br>University of Illinois at Urbana-Champaign<br><br><b>David Molitor</b><br>Associate Professor of Finance<br>Gies College of Business<br>University of Illinois at Urbana-Champaign<br><br><b>Julian Reif</b><br>Associate Professor of Finance<br>Gies College of Business<br>University of Illinois at Urbana-Champaign<br><br><b>George Deltas</b><br>Professor<br>Department of Economics<br>University of Illinois at Urbana-Champaign |                      |

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